

Amazon Prime Video User Analytics

A Descriptive Analytics Report | README

Project

Descriptive analytics on a simulated 100-user, 31-day (July 2026) Amazon Prime Video subscriber dataset, presented through a Power BI dashboard and a 17-page written report.

1. Overview

This repository contains the report, dataset documentation and supporting files for a Business Analytics assignment. The study summarises **what the data shows** (descriptive analytics only, no forecasting) across demographics, content, viewing behaviour, devices, subscription plans and cross-variable relationships.

2. Project Details

Field	Details
Product	Amazon Prime Video (global video-streaming platform)
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Submitted to	Ms. Monica
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Submission date	21 August 2026
BI tool	Power BI

3. Objectives

- Who are the users? (age and gender profile)
- What do they watch? (genre popularity)
- How do they rate content, and does the most-watched genre also rate highest?
- How does daily engagement change across the month?
- What devices do they watch on?
- What plans do they pay for? (Basic / Standard / Premium)
- Which titles get the most attention?
- Which relationships appear only when variables are cross-tabulated? (gender, plan, device, rating)

4. Dataset at a Glance

Attribute	Value
Users (rows)	100
Period	31 days, July 2026
Fields	User_ID, Age, Gender, Plan, Genre, Title, Rating, Device, Hrs
Total watch time	~10,000 hours (100.0 hrs per user on average)
Distinct titles / genres / plans	24 / 8 / 3

Note on data

Real Prime Video subscriber data is proprietary. The dataset is a **simulated** sample built to reproduce the totals and splits of the source Power BI dashboard. See **Dataset.pdf** for the field dictionary and sample rows.

5. Key Findings

Area	Finding
Demographics	Mean/median age 26.5 (SD 5.69); 58% male, 42% female; secondary peak at age 35 (12 users)
Genre	Sci-Fi most watched (17 users); Thriller least (9)
Ratings	Overall 4.00/5; Documentary and Drama highest (4.2+); Comedy lowest (~3.6)
Daily trend	Peak ~610 (Day 3), low ~190 (Day 10); event-driven, not flat
Devices	TV 29.32%, Mobile 24.74%, Tablet 23.80%, Laptop 22.13%
Plans	Standard 36%, Premium 34%, Basic 30%
Titles	'The Boys' and 'Fallout' tied at 7 viewers; long tail of niche titles
Cross-tabs	43% of women on Premium vs 28% of men; 77% of Basic users are male; Premium out-watches other plans on every device (121.0 hrs avg)

6. Repository Structure

```
Amazon-Prime-Video-User-Analytics/  
|-- README.pdf           <- project overview (this file)  
|-- Dataset.pdf          <- data dictionary + 30-row sample  
|-- Ishika_60_DA_Report.pdf <- full 17-page descriptive report  
|-- prime_video_sample.csv <- machine-readable 30-row sample  
`-- (optional) dashboard.pbix <- Power BI dashboard file
```

7. Report Contents

Section	Topic
Fig. 1-2	Age and gender distribution
Fig. 3-4	Genre distribution and average rating by genre
Fig. 5	Daily watch time trend
Fig. 6	Device usage
Fig. 7	Subscription plan distribution
Fig. 8	Top watched titles
Fig. 9-11	Cross-tabulations (plan x gender, rating x gender x genre, device x plan)
Fig. 12	Composite Power BI dashboard (appendix)

8. Next Steps (Descriptive to Predictive)

- Churn forecasting per segment using watch-hour and rating trends
- Anomaly detection on sudden drops in watch time or title ratings
- Recommendation tuning using the popularity-vs-rating gap (Documentary / Drama)
- Cohort and retention analysis by subscription plan

9. References

- Amazon Prime Video User Analytics Dashboard (Power BI, July 2026)
- Simulated Prime Video subscriber dataset (n = 100)
- McKinney, W. (2022). *Python for Data Analysis* (3rd ed.). O'Reilly Media.
- Course materials: Business Analytics - Descriptive Analytics Assignment Brief (2026)

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